

Parametric Equations

MATH 146 – Calculus II

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Parametrized Curves

Parametric Curve

A **parametric curve** is a curve where the (x, y) coordinates are given by two functions $x = f(t)$ and $y = g(t)$ of a third variable t . The equations $x = f(t)$ and $y = g(t)$ are called the **parametric equations** for the curve. The function $\vec{v}(t) = (x(t), y(t))$ is also called a **vector-valued function**.

Example: Parametrized Curves

Example: Sketch the parametric curve given by $x(t) = t^2 - 2$ and $y(t) = t + 2$.

Parametrized Equations for Lines

Example: Find the equation in slope-intercept form ($y = mx + b$) for the line given parametrically by $x(t) = 2t + 3$ and $y(t) = t - 4$.

Idea: Every parametric curve can be expressed as a curve in x and y , provided you can do the algebra...

Example: Parametrized Curves

Example: Find a parametric equation for the line $y = 3x + 2$.

Example: Find a parametric equation for the line $x = 4$.

Example: Parametrized Curves

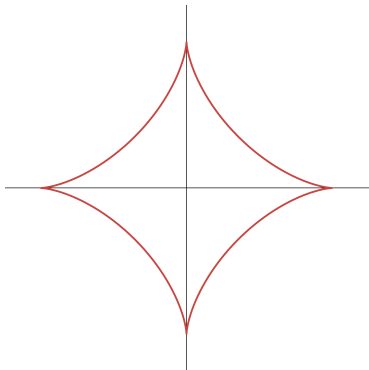
Example: Find a parametric equation for the line containing $(1, 1)$ and $(3, 2)$.

Vector Equation for a Line

Vector Equation for a Line

A **vector equation** for a line is an equation of the form $\vec{v}(t) = \vec{p} + t\vec{d}$, where $\vec{p}$ is a point and $\vec{d}$ is a direction vector.

Derivatives and Tangent Lines for Parametric Curves



Question: How do we find tangents to curves given parametrically?

Idea: We have formulas for $x'(t)$ and $y'(t)$? What do they represent?

Derivatives and Tangent Lines for Parametric Curves

Consider the Chain Rule.

Chain Rule

$$\frac{d}{dx} f(g(x)) = f'(g(x)) g'(x)$$

Take the function $y(t) = y(x(t))$, and derive:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} \quad \implies \quad \frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Example: Derivatives and Tangent Lines for Parametric Curves

Example: Find the equation in slope-intercept form for the line which is tangent to the parametric curve $x(t) = 2t + t^2$ and $y(t) = 7 - 2t$ at $t = 2$.

Example: Find the equation in vector form.

Arc Length in Parametric Coordinates

Theorem (Arc Length in Parametric Coordinates)

Given differentiable parametric functions $x(t)$ and $y(t)$, the arc length of the curve given parametric by x and y over the interval $[a, b]$ is

$$\int_a^b \sqrt{(x'(t))^2 + (y'(t))^2} dt.$$

Example: Arc Length in Parametric Coordinates

Example: Find the circumference of a circle of radius $r = 4$.